Atay Kim

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Mathematics undergraduate combining scientific machine learning, numerical PDEs, full-stack software engineering, and education technology; experienced in taking research and products from experiments to public, reproducible outputs.

Education

Nazarbayev University — Astana, Kazakhstan
BSc Mathematics, School of Computing and Artificial Intelligence (SCAI) — *Aug 2023 - May 2028 (expected)*

- Selected coursework: Real Analysis, Probability, Mathematical Statistics, Linear and Abstract Algebra, Differential Equations, Partial Differential Equations, Numerical Methods, and Programming Fundamentals.

Udacity × Astana Hub — Online
Data Analysis Fundamentals — *Apr 2025 - Aug 2025*

Research Experience

INMET, Nazarbayev University — Astana, Kazakhstan
Research Assistant · Supervisor: Prof. Yanwei Wang — *Jan 2026 - Present*

- Built a PyTorch Fourier Neural Operator benchmark for heterogeneous 1D heat conduction, generating **64-node finite-volume** reference data and evaluating temperature, heat-flux, and discrete PDE-residual errors.
- Compared near-equal **139k-parameter** models with 4 versus 32 Fourier modes; in the hardest out-of-distribution regime, 32 modes reduced median temperature error from **1.170% to 0.709%** and flux error from **17.062% to 11.204%**, with a 6.34 ± 1.14 **percentage-point** flux advantage ($p = 0.03125$).

Ghent Analysis & PDE Center, Ghent University — Ghent, Belgium
Visiting Research Student · Mentor: Dr. Bolys Sabitbek — *Jun 2026 - Jul 2026*

- Generated and standardized **300 synthetic 2D incompressible smoke-flow simulations** with randomized sources and sinks in PhiFlow; implemented and trained a PyTorch FNO on an NVIDIA Tesla T4.
- Evaluated autoregressive trajectories against numerical simulations and documented blur, periodic-boundary artifacts, rollout drift, and distribution shift; released reproducible data-generation, training, evaluation, and demo components.

Publications & Presentations

- **"Fourier-Mode Allocation and Physical Error in a One-Dimensional Heat Problem."** Published abstract, CMAI 2026 Book of Abstracts, p. 31, ISBN 978-601-08-6470-2; presented at CMAI 2026.
- **"Fourier Neural Operators for Engineering PDEs."** Poster and interactive demonstration presented at AIDME 2026; live FNO demo.

Professional Experience

BI Group (Aitanym.kz) — Astana, Kazakhstan
Education Consultant — *Jan 2026 - Present*

- Created **1,500+ mathematics practice problems** and **30+ lessons** for an AI-personalized UNT preparation product, turning curriculum requirements into structured learning content.
- Designed problem sequences across topics and difficulty levels so the product can assign targeted practice while preserving clear statements, reliable solutions, and coherent progression.

Technical Skills

Programming: Python, TypeScript, JavaScript, SQL, HTML/CSS
ML / Scientific: PyTorch, NumPy, pandas, Matplotlib, PhiFlow, Fourier Neural Operators, numerical simulation
Web / Systems: React, Next.js, FastAPI, Node.js, PostgreSQL, Docker, Tailwind CSS, Git/GitHub, Linux
Other: MATLAB, LaTeX, Figma; English (C1, IELTS 7.5), Russian (native)

Software Engineering Projects

CoursePlanner | *TypeScript, React, React Flow, Vite, Tailwind CSS* | Live · Code 2024 - Present

- Built a degree-planning application used by **2,000+ students each semester** (approximately **30% of NU undergraduates**), representing prerequisites as an interactive graph and validating semester plans before registration.

Guide for Applicants | *TypeScript, Quartz* | Live · Code 2022 - Present

- Developed and maintained a searchable university-admissions knowledge base used by **4,000+ applicants**, including the publishing workflow and content architecture for guidance across Kazakhstan, the CIS, and international universities.

Confind · nFactorial AI Cup 2025 | *React, TypeScript, FastAPI, LLaMA 3.1* | Live · Code 2025

- Built a bilingual voice-first accommodation-search prototype using LLaMA 3.1 for query extraction and BeautifulSoup/Selenium for live listing retrieval.

Astra Travel Group | *Responsive web development* | Live 2026

- Designed and developed astratravelgroup.kz end to end, covering information architecture, visual direction, responsive implementation, and production launch.

Leadership

Beyond Curriculum Kazakhstan
Organizing Committee Member *Nov 2023 - Present*

- Organized **five international contests** reaching **2,000+ participants across four countries**; coordinated up to **130** organizers, jury members, designers, and volunteers and managed an event budget above **USD 20,000**.
- Led partner communication with Chevron, Air Astana, Nazarbayev Foundation, Al-Farabi Kazakh National University, Energo University, and Baiterek School.

Olscience Circles Kazakhstan / Remote
Mathematics Track Lead *May 2023 - Nov 2023*

- Led **10 mentors** delivering **30+ lessons** and **100+ hours** of olympiad instruction to **60+ students** across the CIS; helped run two offline olympiad camps.

Teaching

Private Mathematics, Physics, and Computer Science Tutoring Remote / Astana
Self-Employed Tutor *2019 - Present*

- Taught **30+ one-to-one students** and **100+ students in groups**, from Grade 1 through second-year university; prepared students for SAT, NUET, UNT, mathematics olympiads, and university mathematics.
- Achieved outcomes including **SAT Math 590 to 750 in six weeks**, **SAT Math 780 in eight weeks** on a first attempt, **NUET 177 in nine weeks**, admissions to NU/NYUSH/MIPT/MSU/MAI, and a FEMO International Final bronze medal.

Selected Awards

Dean's List · Nazarbayev University

Honorable Mention · Kazakhstan Republican Mathematics Olympiad 2023

Silver Medal, best result from Kazakhstan · International Youth Math Challenge 2022

Absolute 2nd Place · National Mathematics Olympiad of Kyrgyzstan 2021

1st Place · American Mathematics Competition, AMC 10 2021